

Cloud Computing

Lab-6

Name: Monisha Sharma

SRN: PES2UG23CS906

Class: 6F

Date: 19/02/2026

```
mona@MonasLaptop:~/PES2UG23CS906$ docker build -t jenkins-docker -f Dockerfile.jenkins .
[+] Building 151.8s (6/6) FINISHED          docker:default
=> [internal] load build definition from Dockerfile.jenkins 0.0s
=> => transferring dockerfile: 178B 0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins-lts 0.1s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/2] FROM docker.io/jenkins/jenkins-lts@sha256:23b84 0.7s
=> => resolve docker.io/jenkins/jenkins-lts@sha256:23b84 0.1s
=> [2/2] RUN apt-get update && apt-get install -y 120.1s
=> exporting to image 30.5s
=> => exporting layers 26.0s
=> => exporting manifest sha256:a56caaf9ce5f4000f66b8bd2 0.0s
=> => exporting config sha256:62365e0c3400851966d6f365cc 0.0s
=> => exporting attestation manifest sha256:de0327762f0a 0.0s
=> => exporting manifest list sha256:410b805d99e139aca41 0.0s
=> => naming to docker.io/library/jenkins-docker:latest 0.0s
=> => unpacking to docker.io/library/jenkins-docker:late 4.3s
mona@MonasLaptop:~/PES2UG23CS906$ docker logs jenkins
Error response from daemon: No such container: jenkins
mona@MonasLaptop:~/PES2UG23CS906$ docker run -d -p 8080:8080 -p
50000:50000 -v jenkins_home:/var/jenkins_home -v
/var/run/docker.sock:/var/run/docker.sock --name jenkins jenkins
-docker
flag needs an argument: 'v' in -v

Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Run 'docker run --help' for more information
-bash: /var/run/docker.sock:/var/run/docker.sock: No such file o
r directory
mona@MonasLaptop:~/PES2UG23CS906$ docker run -d -p 8080:8080 -p
50000:50000 -v jenkins_home:/var/jenkins_home -v /var/run/docke
r.sock:/var/run/docker.sock --name jenkins jenkins-docker
bfa96b3fd7a11b208fe478787784e1b45a1e09c277ce94a6913b0f34ff025dc1
mona@MonasLaptop:~/PES2UG23CS906$
```

← → ↻ http://localhost:8080

Task List Techjunkie Tools Tip of My Tongue - Ch... Fun school Sem6 chess mar github Amazon Web Services LAB PROGRAM SUBM... Other Bookmarks

Jenkins
build history

Build Queue
No builds in the queue.

Build Executor Status 0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API Jenkins 2.541.2

http://localhost:8080/user/pes2ug23cs906

PES2UG23CS906

- Theme Light
- My Views
- Account
- Appearance
- Preferences
- Security
- Experiments
- Credentials

```
mona@MonasLaptop: ~/PES
2026-02-19 06:59:52.170+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@2aa6311a{HTTP/1.1, (http/1.1){0.0.0.0:8080}}
2026-02-19 06:59:52.170+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@78525ef9{STARTING}[12.1.5,sto=0] @1657ms
2026-02-19 06:59:52.174+0000 [id=40] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-02-19 06:59:52.377+0000 [id=39] INFO jenkins.model.Jenkins#<init>: Starting version 2.541.2
2026-02-19 06:59:52.600+0000 [id=48] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-02-19 06:59:52.619+0000 [id=56] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-02-19 06:59:52.633+0000 [id=52] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-02-19 06:59:53.642+0000 [id=52] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-02-19 06:59:53.643+0000 [id=57] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-02-19 06:59:53.968+0000 [id=57] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-02-19 06:59:53.969+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-02-19 06:59:53.973+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-02-19 06:59:54.057+0000 [id=75] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-02-19 06:59:54.519+0000 [id=55] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-02-19 06:59:54.519+0000 [id=55] INFO jenkins.install.SetupWizard#init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]> *****
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> 26f8b2bdb3da4798a2a97471d2c18a35
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
2026-02-19 07:00:00.474+0000 [id=55] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-02-19 07:00:00.515+0000 [id=39] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-02-19 07:00:02.787+0000 [id=75] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.task
s.Maven.MavenInstaller
```

TASK2

Browser address bar: <http://localhost:8080/job/PES2UG23CS906-backend-build/4/console>

Jenkins / PES2UG23CS906-backend... / #4

```
#9 [5/5] RUN g++ app.cpp -o app
#9 DONE 0.6s

#10 exporting to image
#10 exporting layers
#10 exporting layers 18.2s done
#10 exporting manifest sha256:e30ab78ff5e5571ef62b177ee42b61a6798f97fb0d814e9884d637236e99971c 0.1s done
#10 exporting config sha256:d54320873ed7adfd5981f7a961768cfd0898cccafd1e35de3384b1277d54 0.0s done
#10 exporting attestation manifest sha256:c9a412821aa3efff169a8aace40133284bc869f6d9515371a986f9417b87f7bb
#10 exporting attestation manifest sha256:c9a412821aa3efff169a8aace40133284bc869f6d9515371a986f9417b87f7bb
0.1s done
#10 exporting manifest list sha256:01cbb4c377047e18e26904593fd30abeddf92ed755c6be9ece77f842ac4bfe7 0.0s done
#10 naming to docker.io/library/backend-app:latest
#10 naming to docker.io/library/backend-app:latest done
#10 unpacking to docker.io/library/backend-app:latest
#10 unpacking to docker.io/library/backend-app:latest 5.2s done
#10 DONE 23.8s
Finished: SUCCESS
```

Jenkins 2.541.2

Browser address bar: <http://localhost:8080/job/PES2UG23CS906-backend-build/>

Jenkins / PES2UG23CS906-backend...

Status

Changes

Workspace

Build Now

Configure

Delete Project

Git Polling Log

Rename

✓ PES2UG23CS906-backend-build

Permalinks

- [Last build \(#4\), 4 min 15 sec ago](#)
- [Last stable build \(#4\), 4 min 15 sec ago](#)
- [Last successful build \(#4\), 4 min 15 sec ago](#)
- [Last failed build \(#3\), 5 min 14 sec ago](#)
- [Last unsuccessful build \(#3\), 5 min 14 sec ago](#)
- [Last completed build \(#4\), 4 min 15 sec ago](#)

Builds >

Filter

Today

- ✓ #4 7:30 AM
- ✗ #3 7:29 AM
- ✗ #2 7:26 AM

Task3

← → ↻ http://localhost:8080/job/PES2UG23CS906-backend-build/build?delay=0sec

Task List Techjunkie Tools Tip of My Tongue - Ch... Fun school Sem6 chess mar github Amazon Web Services PES2UG23CS906-back... Other Bookmarks

Jenkins / PES2UG23CS906-backend...

- Status
- Changes
- Workspace
- Build with Parameters
- Configure
- Delete Project
- Git Polling Log
- Rename

Project PES2UG23CS906-backend-build

This build requires parameters:

BACKEND_COUNT
Number of backend containers

1

Build Cancel

Builds > Filter

Jenkins / PES2UG23CS906-backend... / #6

```
#10 exporting to image
#10 exporting layers done
#10 exporting manifest sha256:e30ab78ff5e5571ef62b177ee42b61a6798f97fb0d814e9084d637236e99971c done
#10 exporting config sha256:d54320873ed7adfd5981f7a961768cfd0898ccecacf1e35de3384b1277d54 done
#10 exporting attestation manifest sha256:65003d9ea2fb82f340dc94035f9e2e9e1f79a7b57eff7ba3d667123048018170
0.1s done
#10 exporting manifest list sha256:479f3e8a504242b559911a3a02f734cee7b7b880c789b9f9adfc506f40aa6c34
#10 exporting manifest list sha256:479f3e8a504242b559911a3a02f734cee7b7b880c789b9f9adfc506f40aa6c34 0.0s done
#10 naming to docker.io/library/backend-app:latest done
#10 unpacking to docker.io/library/backend-app:latest 0.0s done
#10 DONE 0.4s
+ docker rm -f backend1 backend2
Error response from daemon: No such container: backend1
Error response from daemon: No such container: backend2
+ [ 1 = 1 ]
+ docker run -d --name backend1 backend-app
ea3434c2dcb97d00e531944b3a08aa867eb2e1c4d683245fed38e3010ff8fbf9
Finished: SUCCESS
```

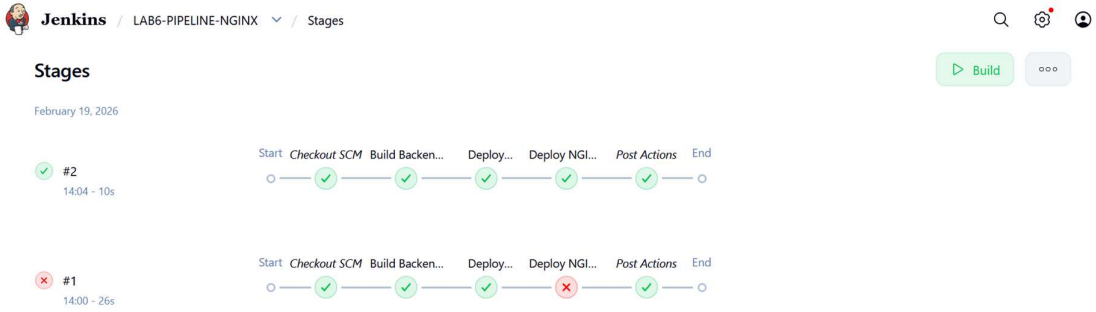
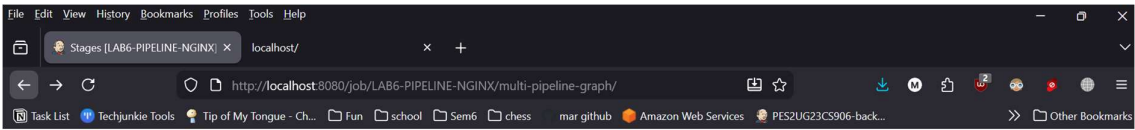
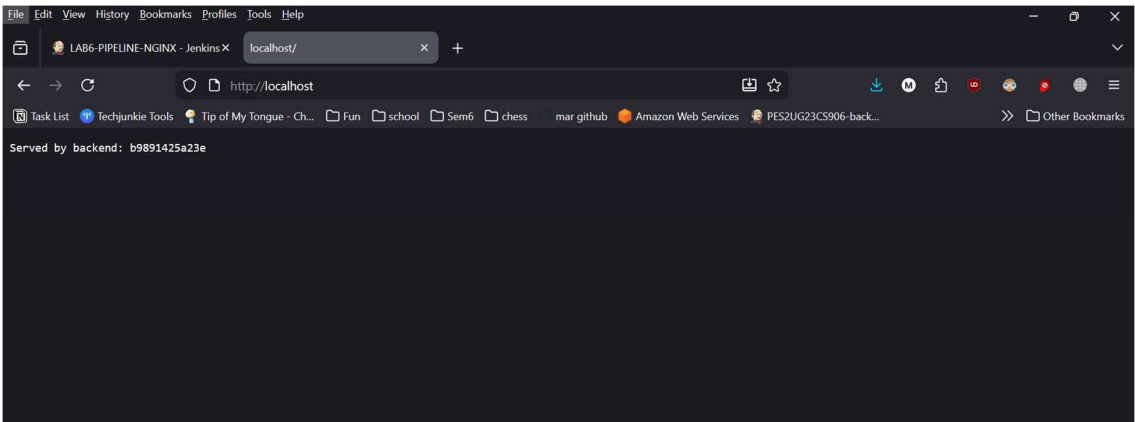
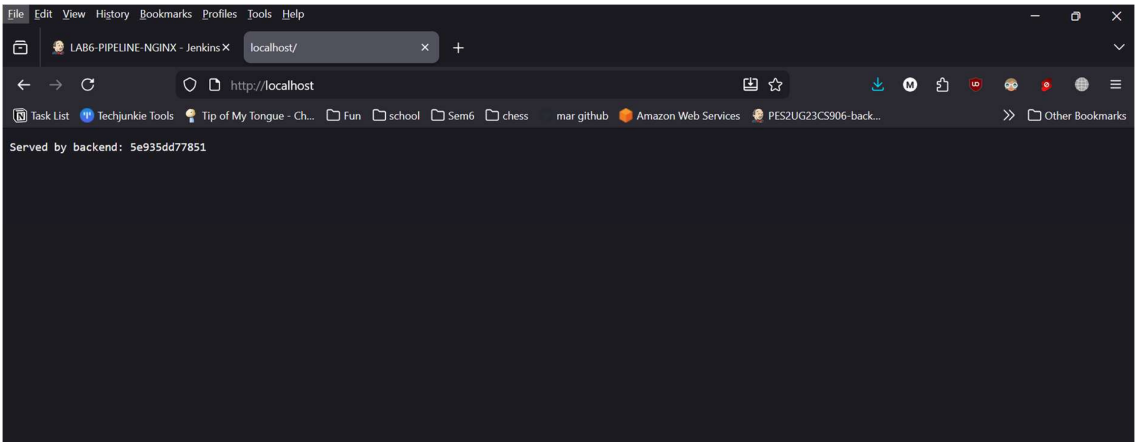
← → ↻ http://localhost:8080/job/PES2UG23CS906-backend-build/77/console

Task List Techjunkie Tools Tip of My Tongue - Ch... Fun school Sem6 chess mar github Amazon Web Services PES2UG23CS906-back... Other Bookmarks

Jenkins / PES2UG23CS906-backend... / #7

```
#10 exporting manifest sha256:e30ab78ff5e5571ef62b177ee42b61a6798f97fb0d814e9084d637236e99971c done
#10 exporting config sha256:d54320873ed7adfd5981f7a961768cfd0898ccecacf1e35de3384b1277d54 done
#10 exporting attestation manifest sha256:8db8c540484d09607bb7965a7f763e9fb640d67cc9ee8cbcec710f53fa3b32b
0.0s done
#10 exporting manifest list sha256:119c7e7f916992ee040d07fd8ad03357a1203289325d826f43caea7ad3f05ce1
#10 exporting manifest list sha256:119c7e7f916992ee040d07fd8ad03357a1203289325d826f43caea7ad3f05ce1 0.0s done
#10 naming to docker.io/library/backend-app:latest done
#10 unpacking to docker.io/library/backend-app:latest done
#10 DONE 0.1s
+ docker rm -f backend1 backend2
backend1
Error response from daemon: No such container: backend2
+ [ 2 = 1 ]
+ docker run -d --name backend1 backend-app
b0c26f4e4adc9da15416fc5d684fac251a0fac8103658e3854a505fa02bbfb95
+ docker run -d --name backend2 backend-app
8b2a86bdfb2992a6937c47043f305f8d5512df9a4d9b04394acc0ee04f6a08e5
Finished: SUCCESS
```

Task4



File Edit View History Bookmarks Profiles Tools Help

Console - #2 - Jenkins localhost/

http://localhost:8080/job/LAB6-PIPELINE-NGINX/2/console

Task List Techjunkie Tools Tip of My Tongue - Ch... Fun school Sem6 chess mar github Amazon Web Services PES2UG23CS906-back...

Jenkins / LAB6-PIPELINE-NGINX #2

```
8265a06bc047bd4cb80c7322e9fbb556e53be26261ba4bef13f191dd235399e
+ docker cp CC_LAB-6/nginx/default.conf nginx-lb:/etc/nginx/conf.d/default.conf
+ docker exec nginx-lb nginx -s reload
2026/02/19 08:35:00 [notice] 36#36: signal process started
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline executed successfully. NGINX load balancer is running.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.2

Task5

Without ip

File Edit View History Bookmarks Profiles Tools Help

LAB6-PIPELINE-NGINX - JenkinsX localhost/

http://localhost

Task List Techjunkie Tools Tip of My Tongue - Ch... Fun school Sem6 chess mar github Amazon Web Services PES2UG23CS906-back... Other Bookmarks

Served by backend: 5e935dd77851

File Edit View History Bookmarks Profiles Tools Help

LAB6-PIPELINE-NGINX - JenkinsX localhost/

http://localhost

Task List Techjunkie Tools Tip of My Tongue - Ch... Fun school Sem6 chess mar github Amazon Web Services PES2UG23CS906-back... Other Bookmarks

Served by backend: b9891425a23e

with

